GASTROINTESTINAL DRUGS

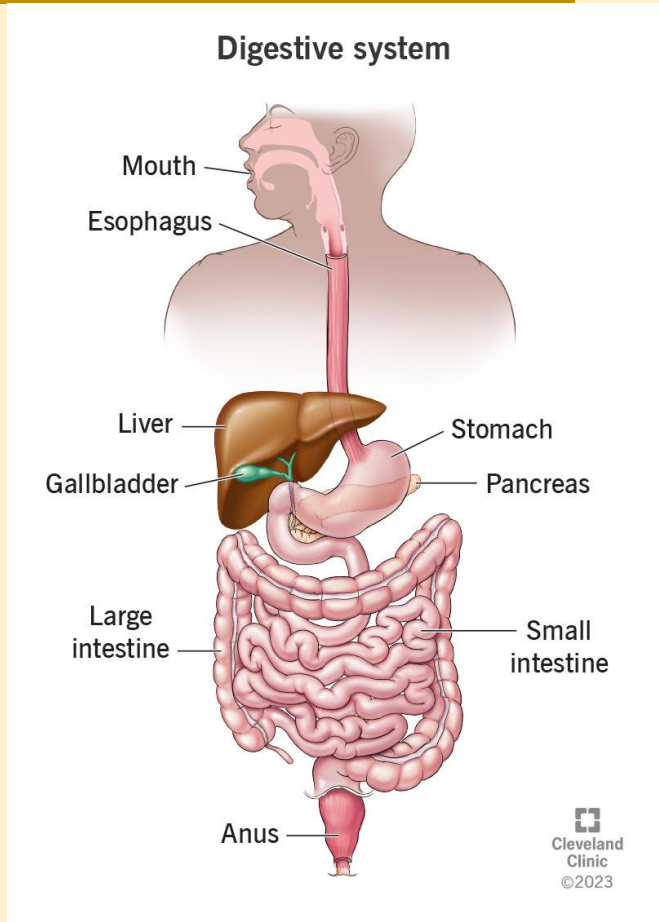
Isaac Edron J. Orig, RMT, MD
APMed-1

GI Tract

Components

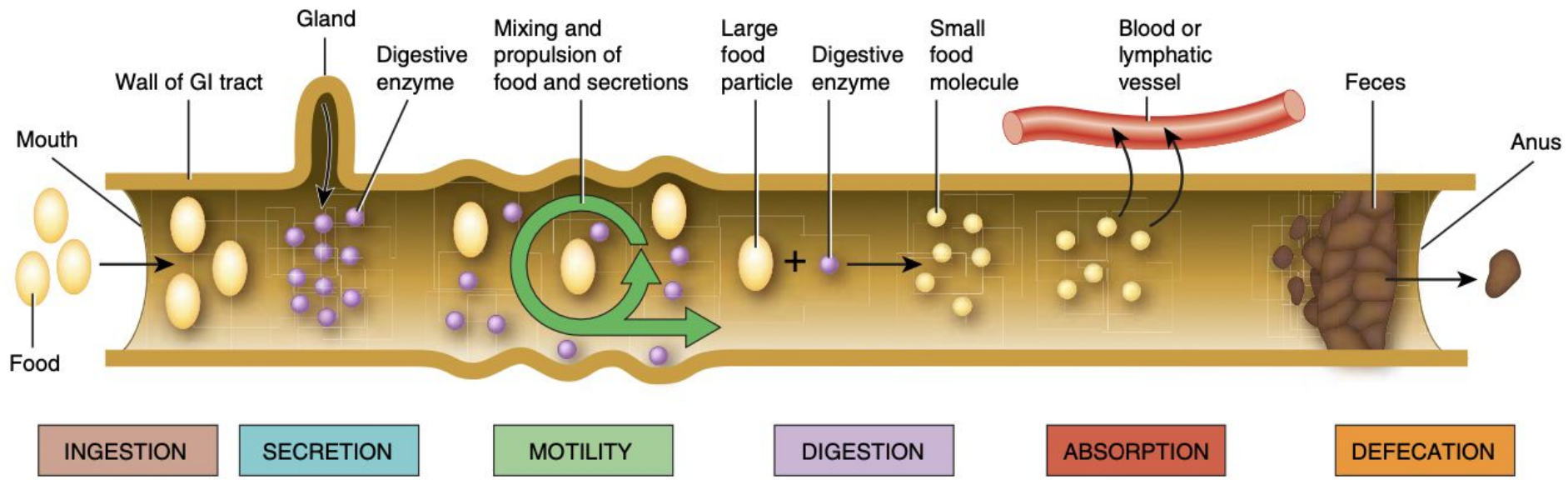
- **Gastrointestinal tract (Alimentary Canal)**
  - Mouth
  - Pharynx
  - Esophagus
  - Stomach
  - Small Intestine
  - Large Intestine

- **Accessory digestive organs**
  - Teeth
  - Tongue
  - Salivary glands
  - Liver
  - Gallbladder
  - Pancreas



Functions

- Ingestion
- Secretion
- Motility
- Digestion
- Absorption
- Defecation

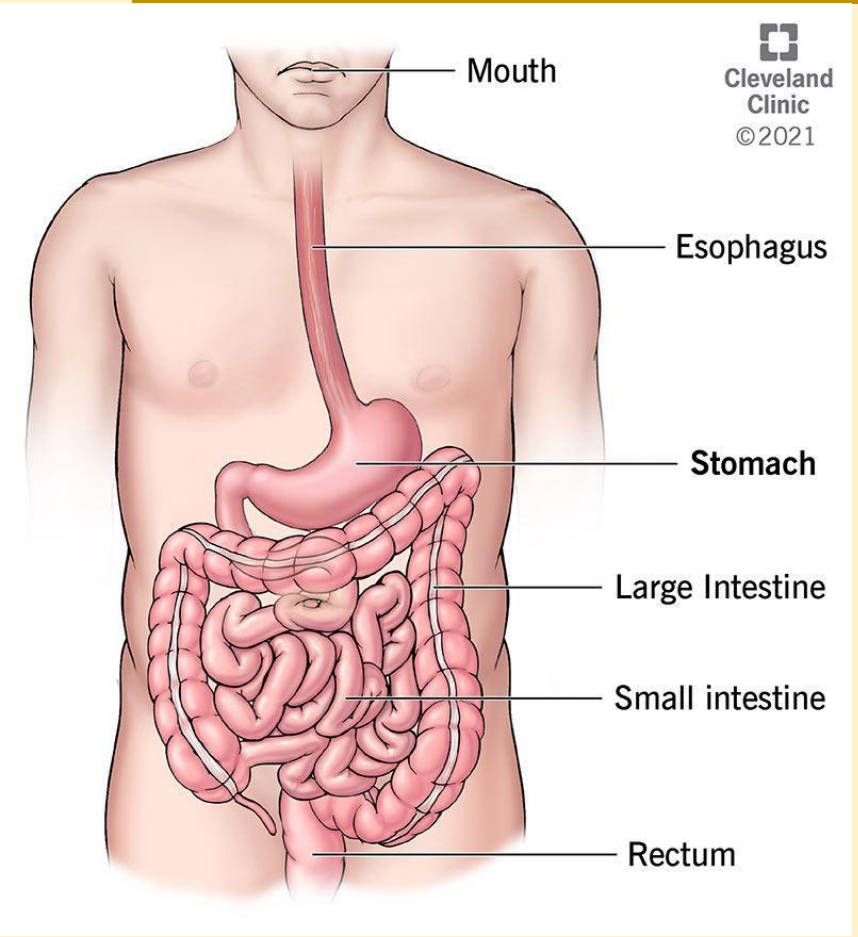


Functions

- **Ingestion:** involves taking foods and liquids into the mouth *(eating)*
- **Secretion:** walls of the GI tract and the accessory digestive organs secrete water, acid, buffer and enzymes into the lumen on the tract
- **Motility:** capability of the GI tract mix and move material along its length
- **Digestion:** process of breaking down ingested food
- **Absorption:** movement of the products of digestion from the lumen of the GI tract into the blood or lymph
- **Defecation:** digested materials that were not absorbed leave the body through the anus

Stomach

STOMACH



- J-shaped enlargement of the GI tract directly inferior to the diaphragm in the abdomen
- Connects the esophagus to the duodenum (1st part of the Small Intestine)
- Most distensible part of the GI tract

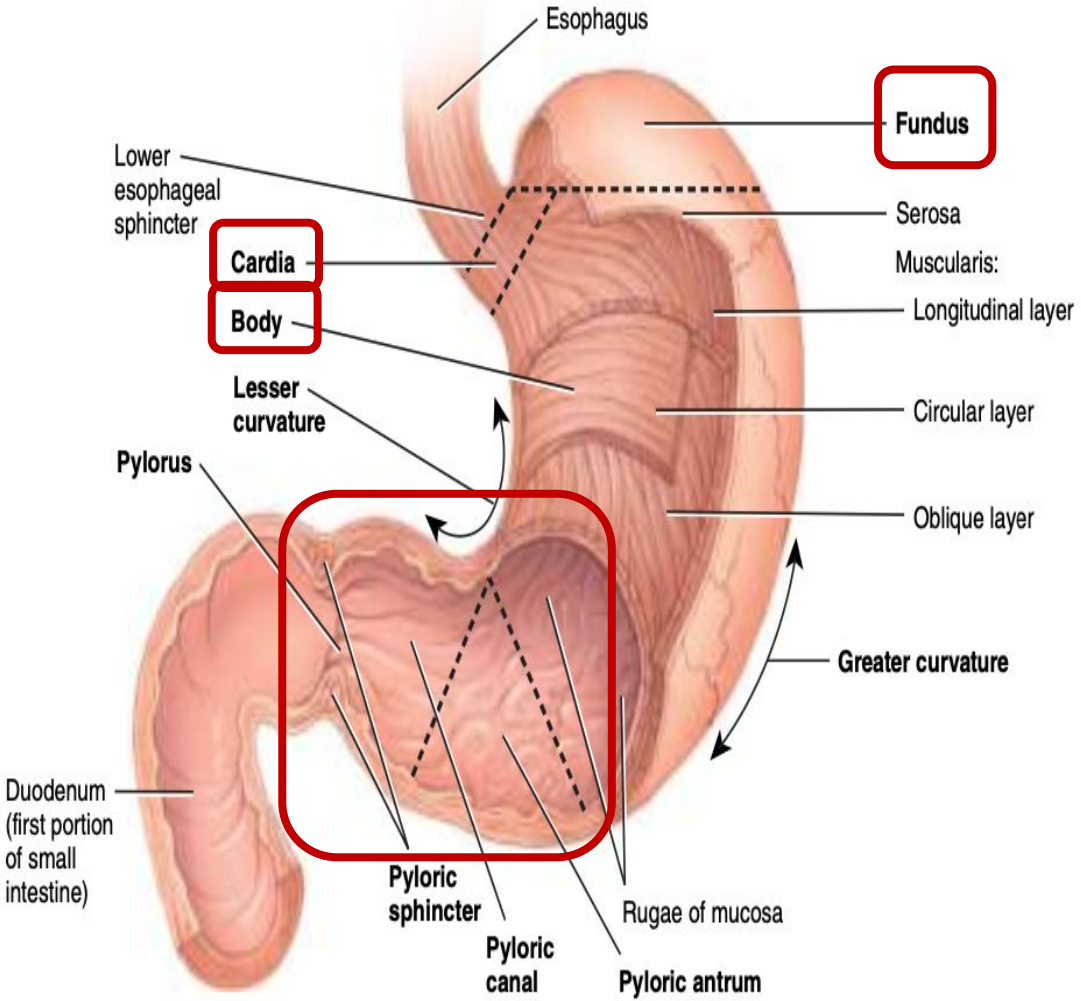
Function:

- Mixing chamber
- Food reservoir
- Secretion (Gastric juice, Gastrin)
- Digestion of starch continues

STOMACH

4 Regions

- **Cardia** - surrounds the opening of the esophagus into the stomach
- **Fundus** – rounded portion superior to and to the left of the cardia
- **Body** – large central portion inferior to the fundus
- **Pylorus** - has 3 regions:
  - **Pyloric antrum** – connects the body of the stomach
  - **Pyloric Canal** – leads to the pylorus (3rd region)
  - **Pylorus** – connects to the duodenum
- **Pyloric sphincter** – muscle sphincter connecting the pylorus to the duodenum

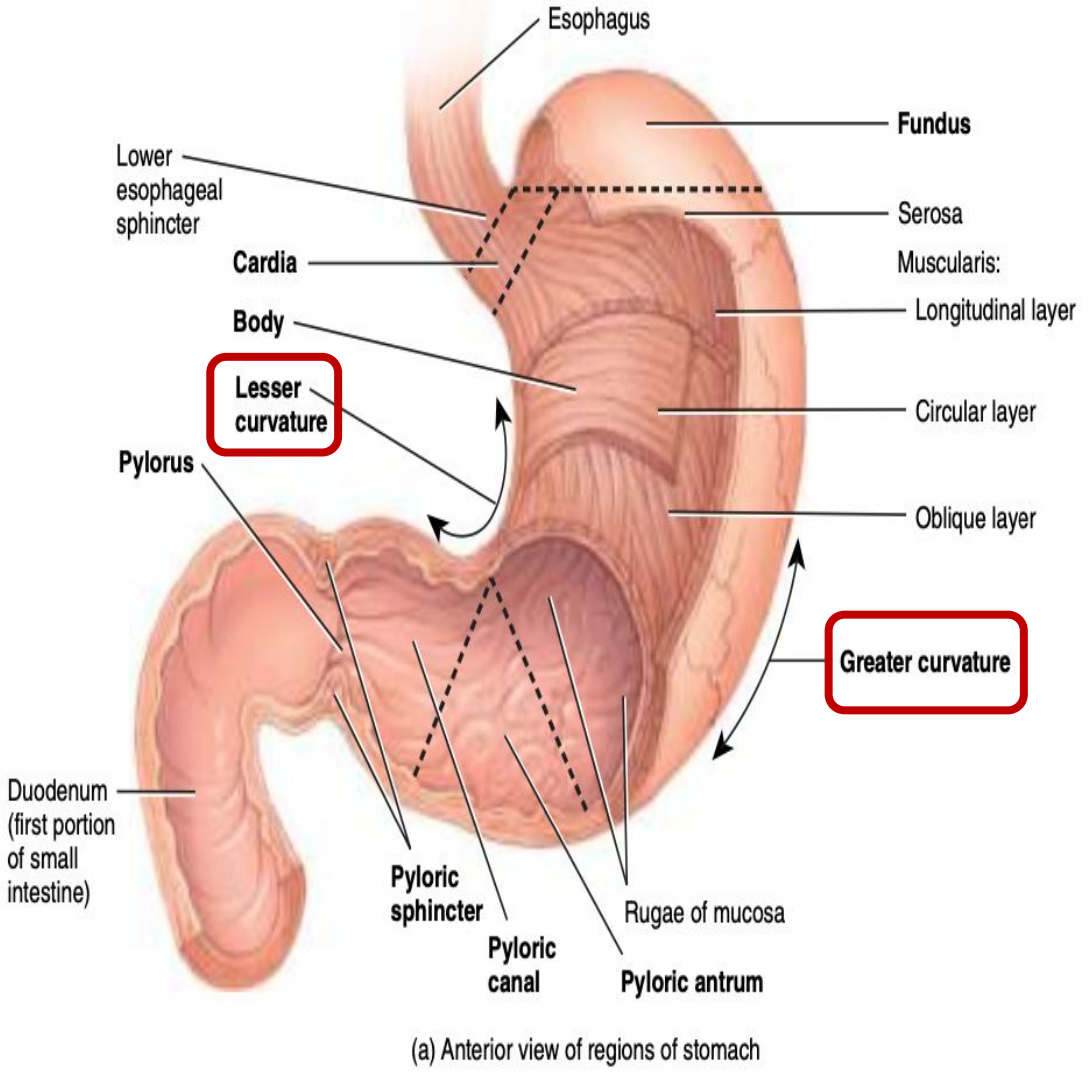


(a) Anterior view of regions of stomach

STOMACH

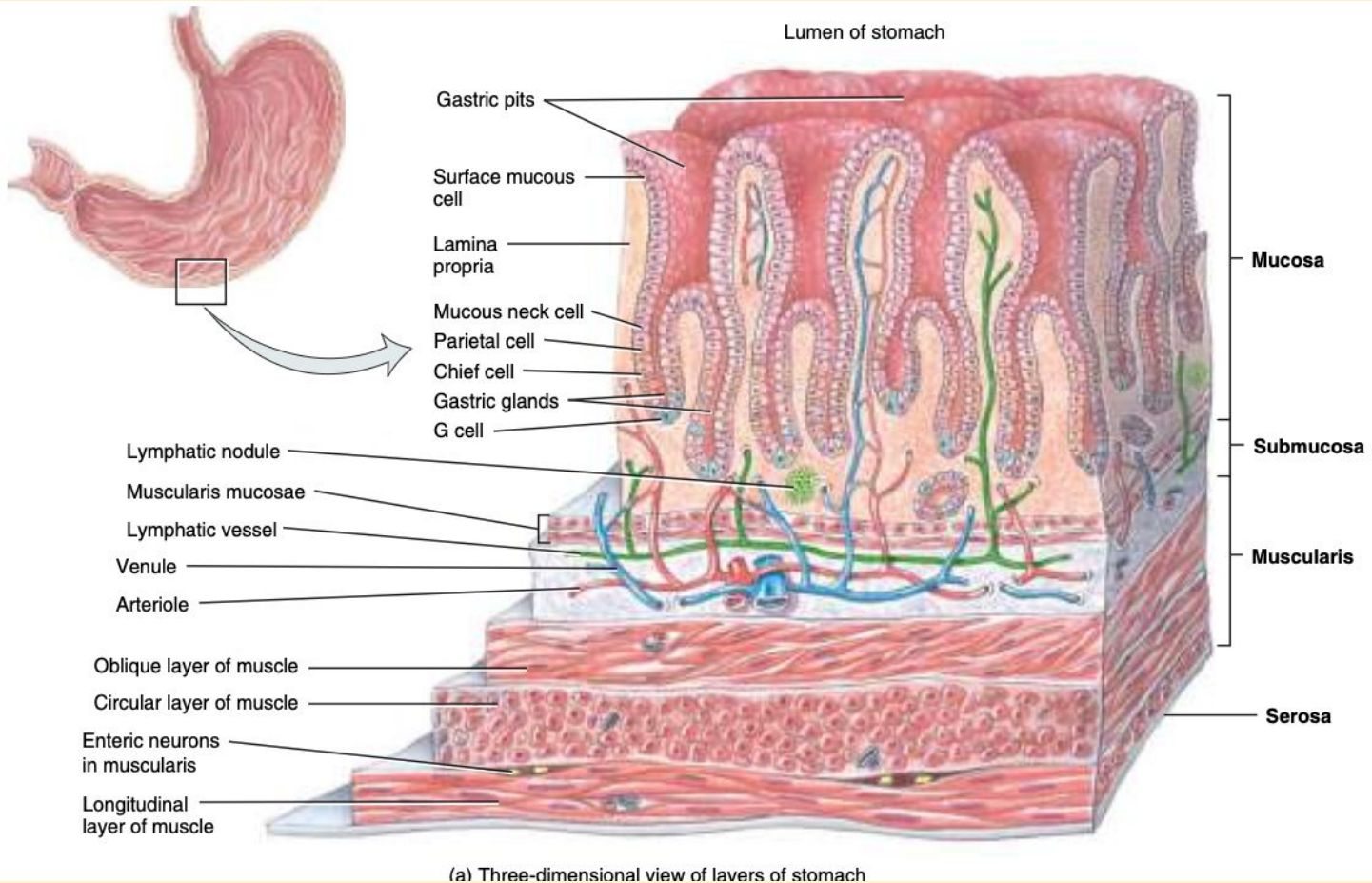
Curves / Borders

- **Lesser curvature** – concave medial border of the stomach
- **Greater curvature** – convex lateral border of the stomach



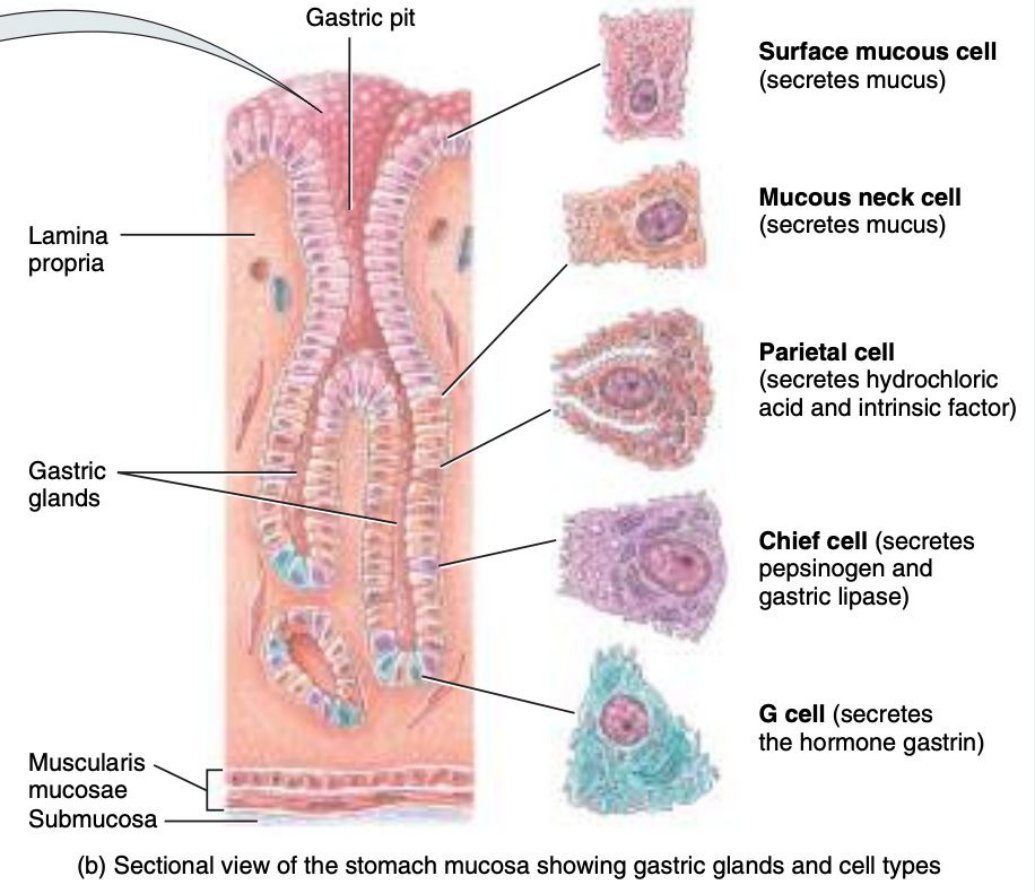
(a) Anterior view of regions of stomach

STOMACH



(a) Three-dimensional view of layers of stomach

STOMACH



(b) Sectional view of the stomach mucosa showing gastric glands and cell types

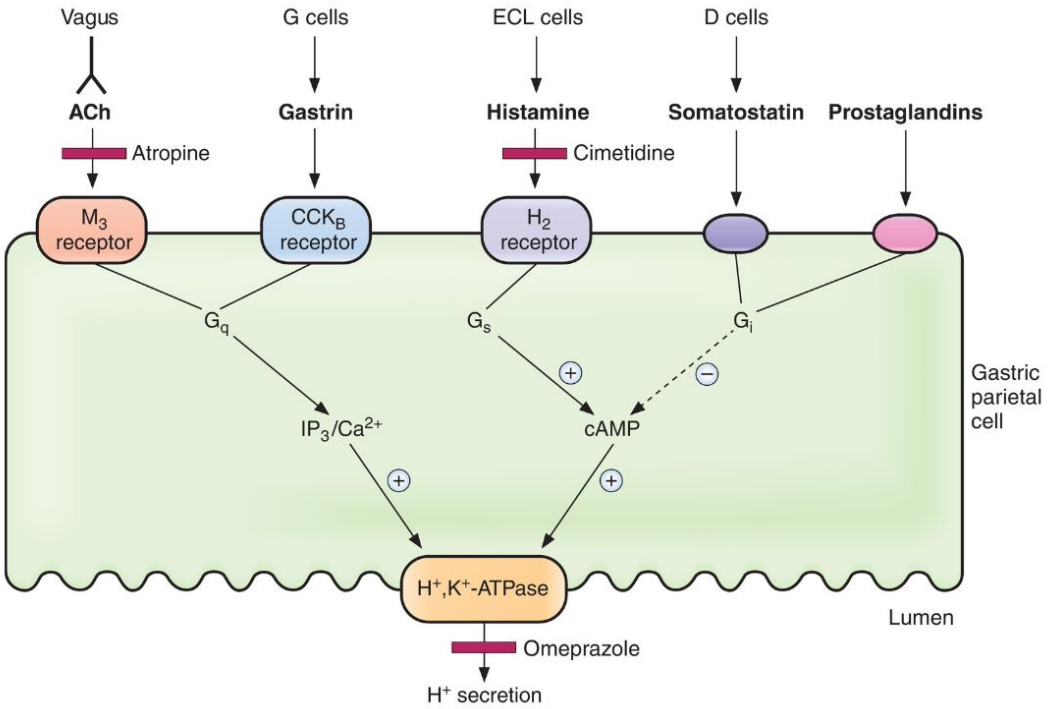
STRUCTURE	ACTIVITY	RESULT
Mucosa		
Surface mucous cells and mucous neck cells	Secrete mucus.	Forms protective barrier that prevents digestion of stomach wall.
	Absorption.	Small quantity of water, ions, short-chain fatty acids, and some drugs enter bloodstream.
Parietal cells	Secrete intrinsic factor.	Needed for absorption of vitamin B_{12} (used in red blood cell formation, or erythropoiesis).
	Secrete hydrochloric acid.	Kills microbes in food; denatures proteins; converts pepsinogen into pepsin.
Chief cells	Secrete pepsinogen.	Pepsin (activated form) breaks down proteins into peptides.
	Secrete gastric lipase.	Splits triglycerides into fatty acids and monoglycerides.
G cells	Secrete gastrin.	Stimulates parietal cells to secrete HCl and chief cells to secrete pepsinogen; contracts lower esophageal sphincter, increases motility of stomach, and relaxes pyloric sphincter.

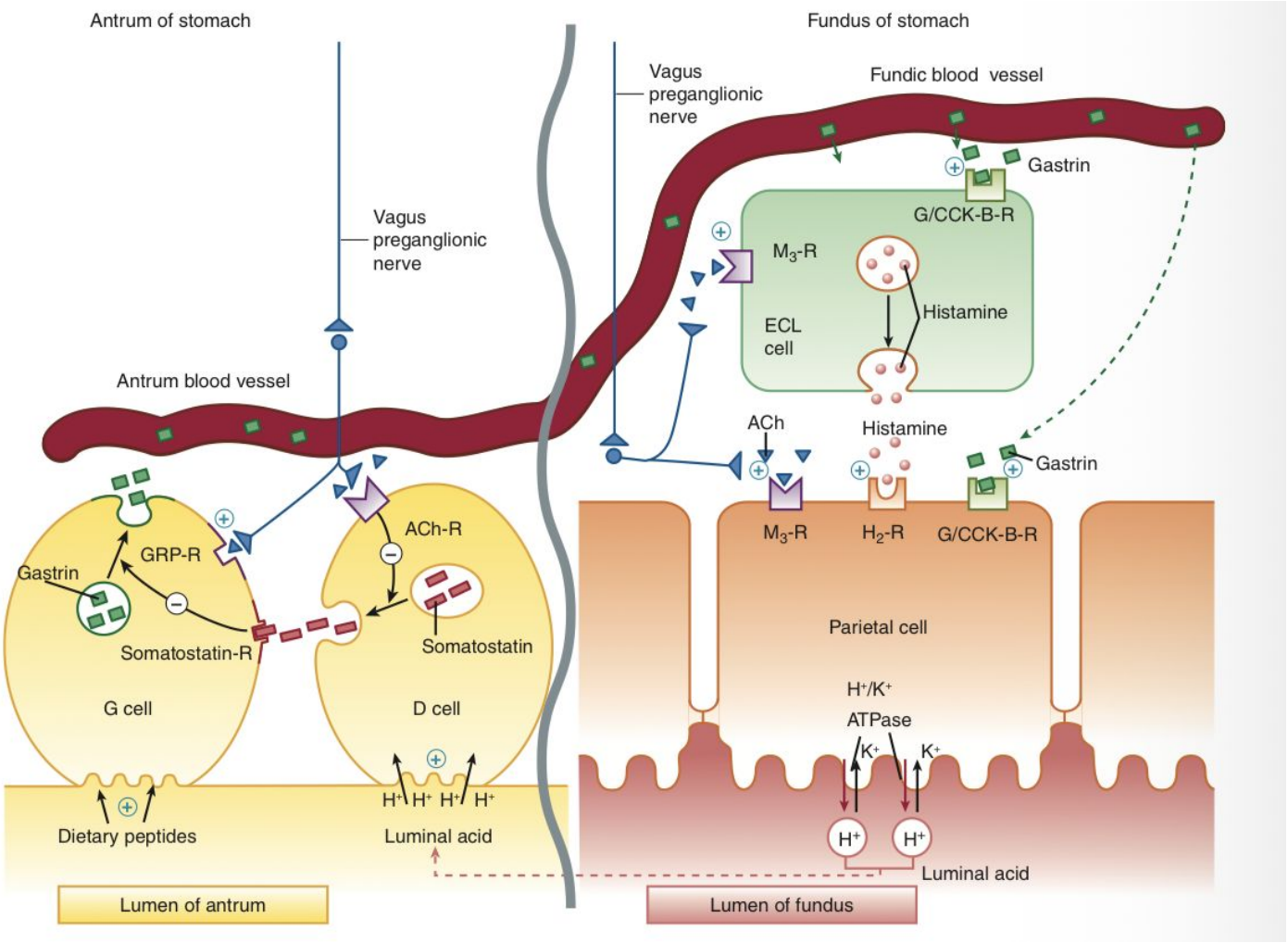
Acid Secretion Physiology

Parietal cell receptors:

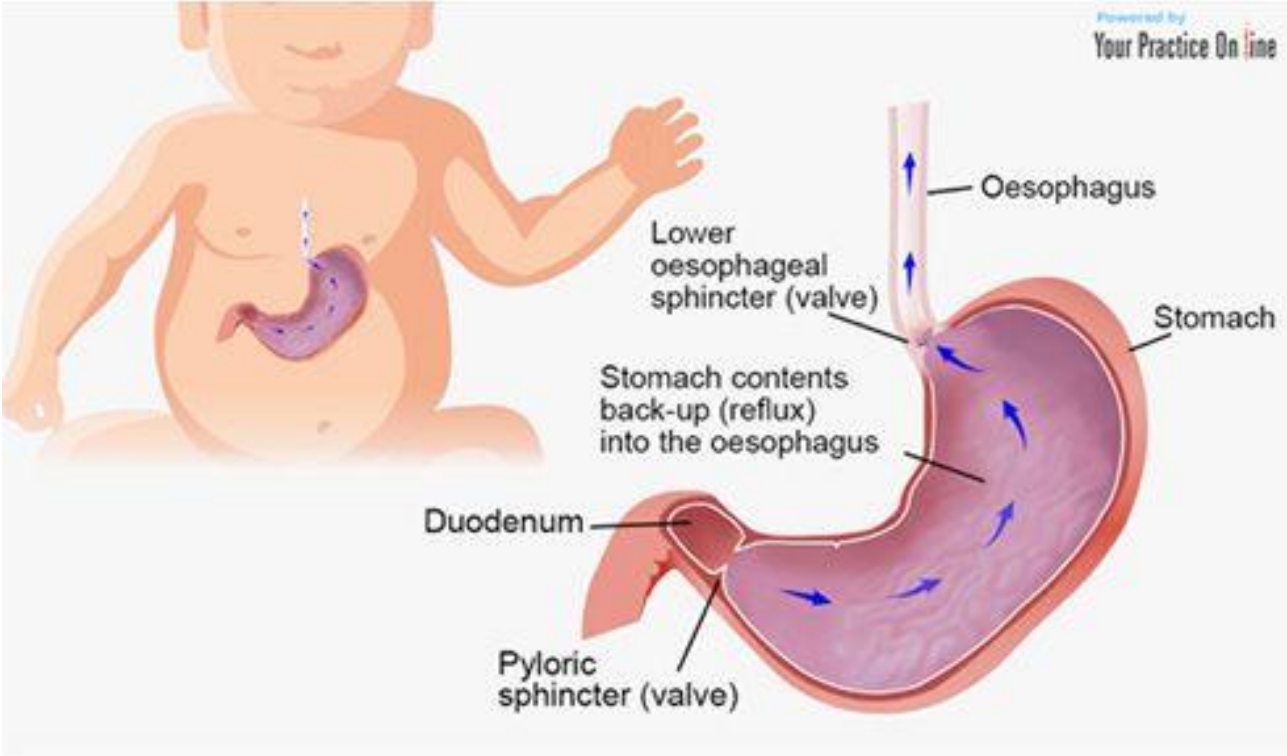
- Gastrin (CCK-B)
- Histamine (H2)
- Acetylcholine (M3)

Stimulates acid secretion from the **proton pump (H/K-ATPase)**

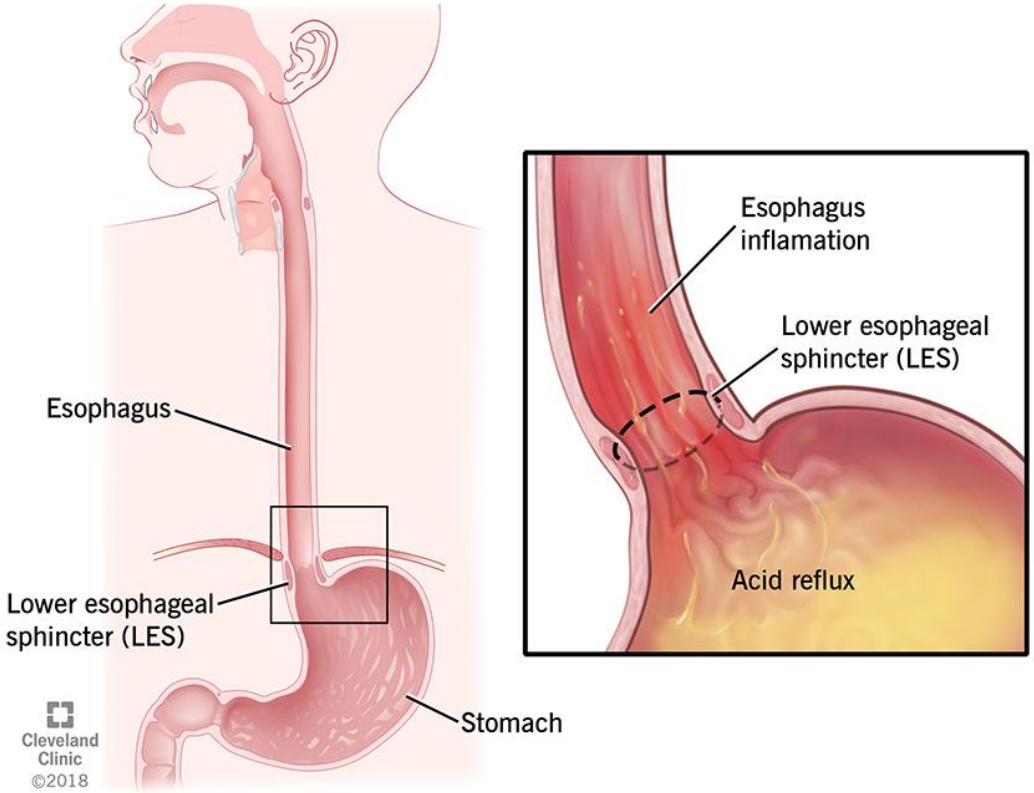



Antrum of stomach
Fundus of stomach
Vagus preganglionic nerve
Vagus preganglionic nerve
Fundic blood vessel
Gastrin
G/CCK-B-R
M3-R
ECL cell
Histamine
Antrum blood vessel
ACh
Histamine
Gastrin
M3-R
H2-R
G/CCK-B-R
GRP-R
Gastrin
ACh-R
Somatostatin-R
Somatostatin
Parietal cell
G cell
D cell
H+/K+ ATPase
K+
K+
H+ H+ H+ H+
H+
H+
Dietary peptides
Luminal acid
Luminal acid
Lumen of antrum
Lumen of fundus

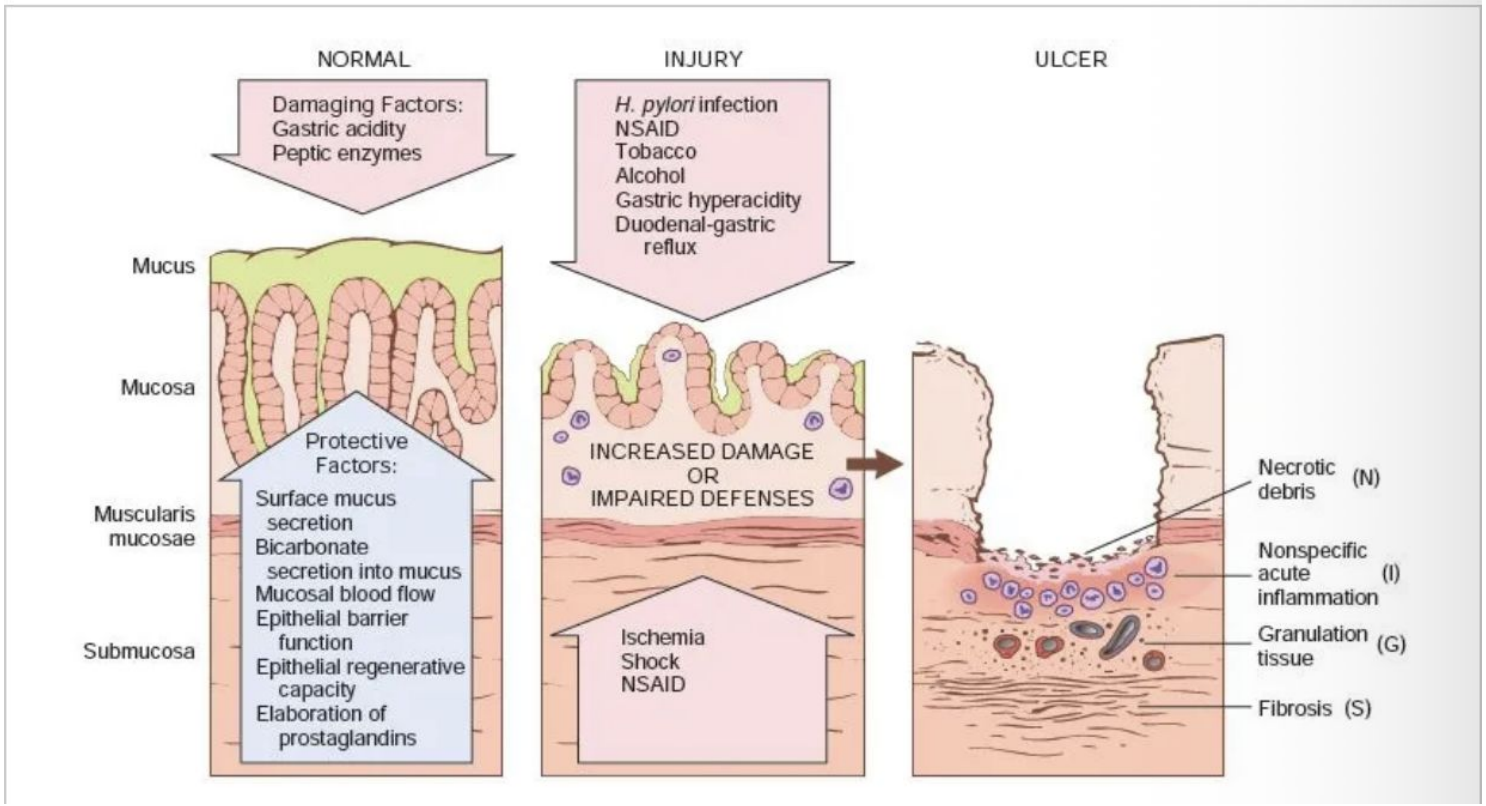
Gastroesophageal Reflux Disease (GERD)



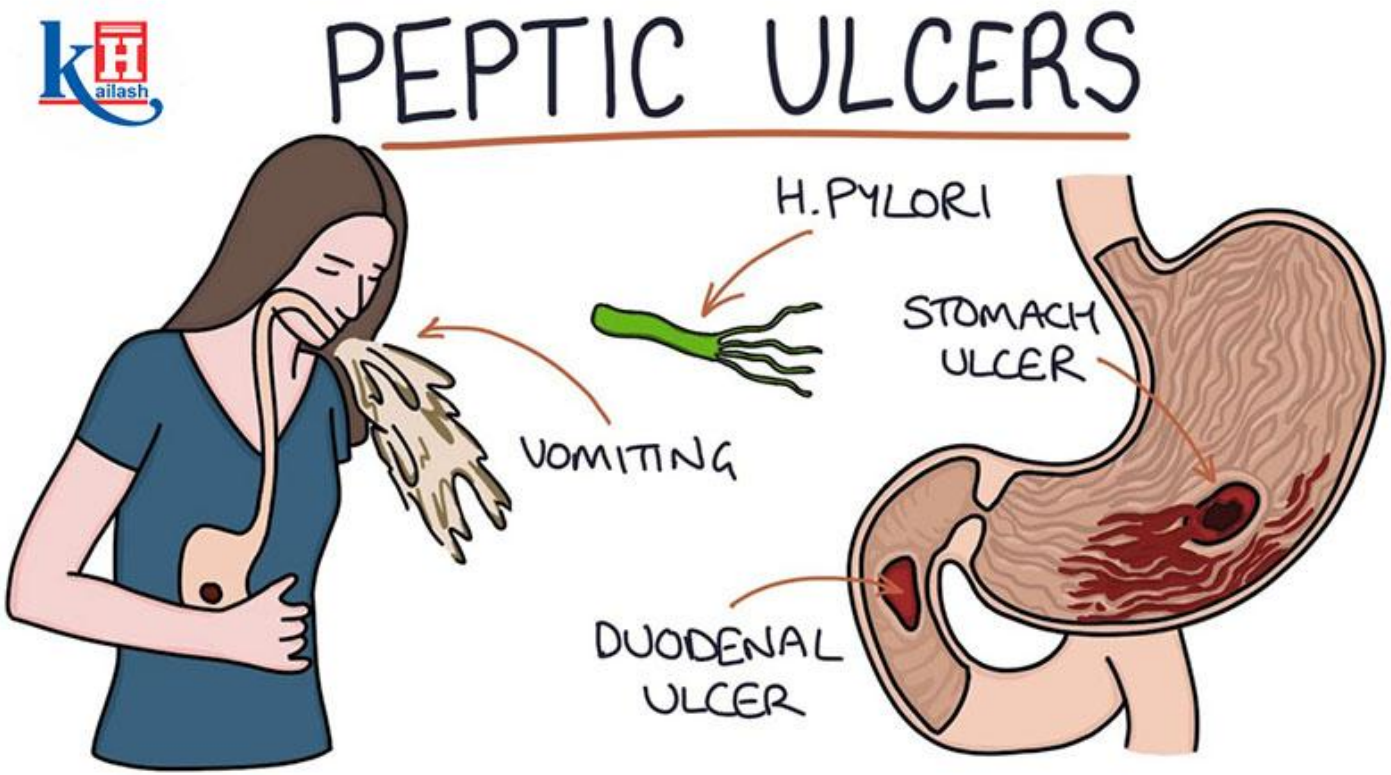
Gastroesophageal Reflux Disease (GERD)

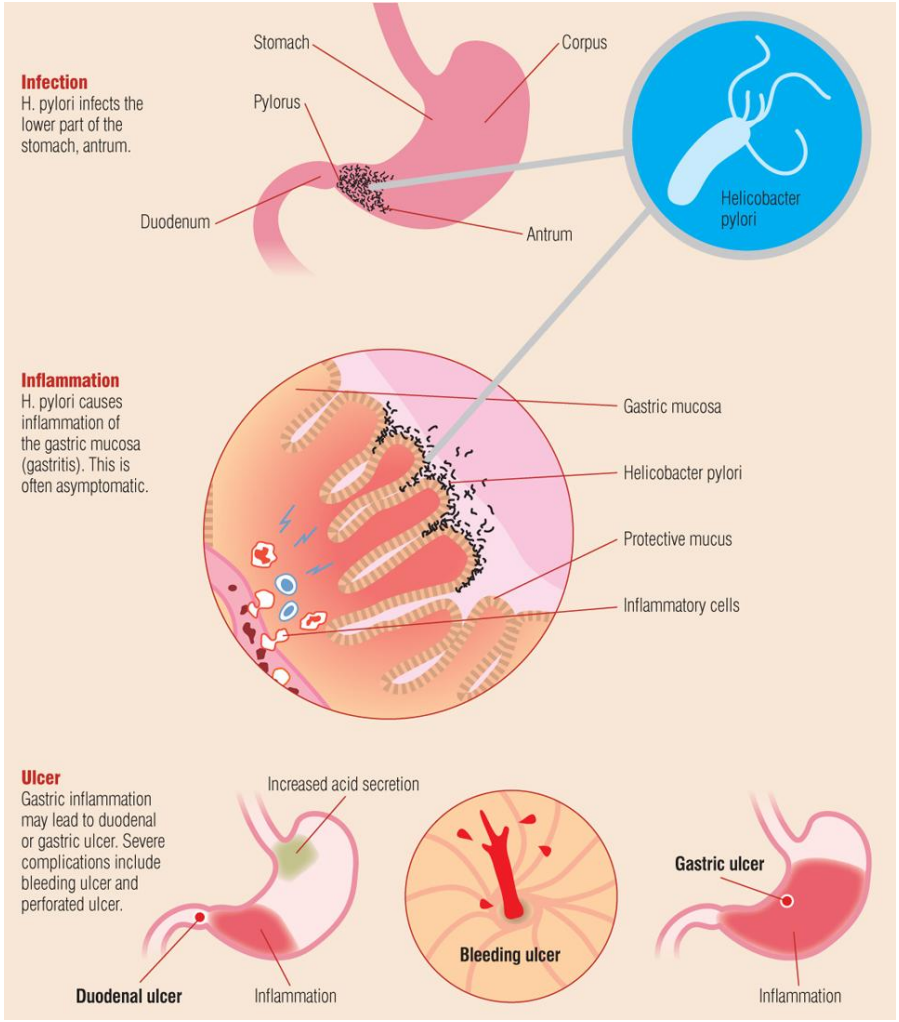


Peptic Ulcer Disease



Peptic Ulcer Disease




Infection
H. pylori infects the lower part of the stomach, antrum.
Stomach
Corpus
Pylorus
Duodenum
Antrum
Helicobacter pylori
Inflammation
H. pylori causes inflammation of the gastric mucosa (gastritis). This is often asymptomatic.
Gastric mucosa
Helicobacter pylori
Protective mucus
Inflammatory cells
Ulcer
Gastric inflammation may lead to duodenal or gastric ulcer. Severe complications include bleeding ulcer and perforated ulcer.
Increased acid secretion
Duodenal ulcer
Inflammation
Bleeding ulcer
Gastric ulcer
Inflammation

Drugs for Acid-Peptic Diseases

Drug Classes for Acid-Peptic Diseases

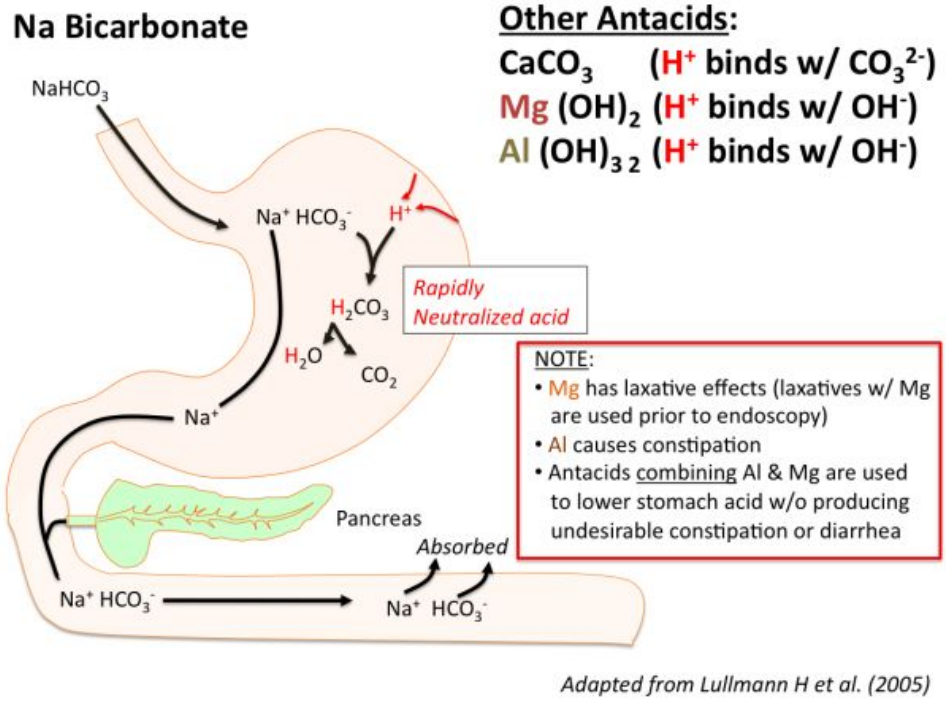
1. Antacids
2. H2-receptor Antagonists
3. Proton Pump Inhibitors (PPI)
4. Sucralfate
5. Prostaglandin Analogs
6. Bismuth Compounds

Antacids

Non-prescription remedies for intermittent heartburn and dyspepsia

1. Sodium bicarbonate
2. Calcium carbonate
3. Magnesium hydroxide/Aluminum hydroxide

A single dose of **156 mEq** antacid given **1 hr** after a meal neutralizes gastric acid for up to **2 hours**



Adapted from Lullmann H et al. (2005)

Sodium Bicarbonate

Aka **Baking Soda**

Formation of **CO2** results in gastric distention and belching

Can cause **metabolic alkalosis**



Calcium carbonate

Reacts more slowly than $NaHCO_3$

May also cause belching or metabolic acidosis

Na and Ca HCO_3 + calcium-containing dairy products = **milk-alkali syndrome**

- hypercalcemia, renal insufficiency, and metabolic acidosis



Magnesium Hydroxide/Aluminum Hydroxide

No gas is generated, no belching

Magnesium = diarrhea

Aluminum = constipation



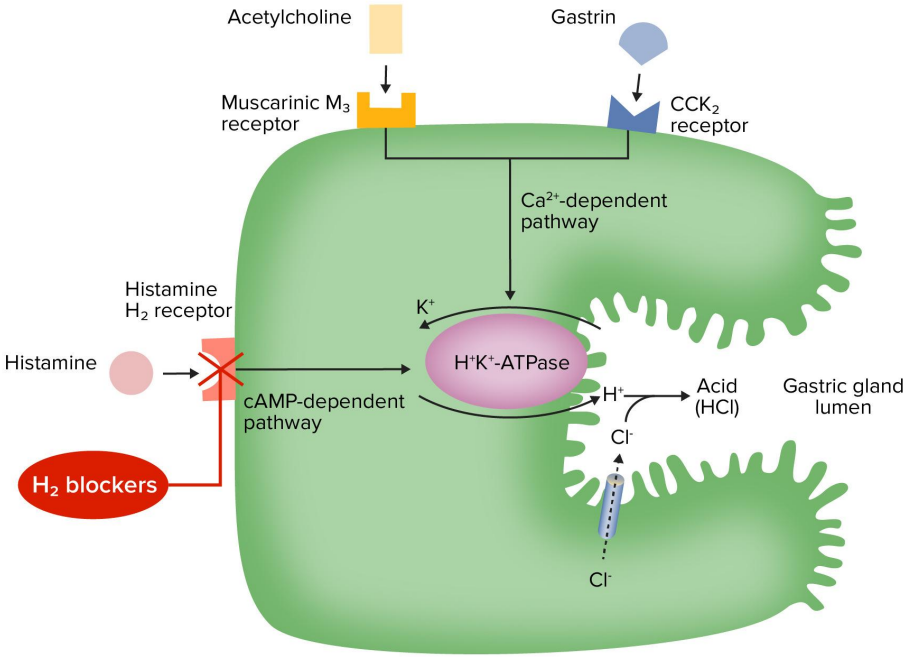
H2 Receptor Antagonists (H2RAs)

Exhibit competitive inhibition at the parietal H2 receptor

Suppresses basal and meal-stimulated acid

Do not affect H1 or H3 receptors

1. Cimetidine
2. Ranitidine
3. Famotidine
4. Nizatidine
    - least first-pass metabolism



Drug	Relative Potency	Dose to Achieve >50% Acid Inhibition for 10 Hours	Usual Dose for Acute Duodenal or Gastric Ulcer	Usual Dose for Gastroesophageal Reflux Disease	Usual Dose for Prevention of Stress-Related Bleeding
Cimetidine	1	400–800 mg	800 mg HS or 400 mg bid	800 mg bid	50 mg/h continuous infusion
Ranitidine	4–10	150 mg	300 mg HS or 150 mg bid	150 mg bid	6.25 mg/h continuous infusion or 50 mg IV every 6–8 h
Nizatidine	4–10	150 mg	300 mg HS or 150 mg bid	150 mg bid	Not available
Famotidine	20–50	20 mg	40 mg HS or 20 mg bid	20 mg bid	20 mg IV every 12 h

bid, twice daily; HS, bedtime.

H2 Receptor Antagonists (H2RAs)

Clinical Uses:

1. **GERD**
   - infrequent heartburn or dyspepsia (<3x per week)
2. **Peptic Ulcer Disease**
   - largely replaced by PPIs
3. **Nonulcer dyspepsia**
4. **Prevention of bleeding from stress-related gastritis**



H2 Receptor Antagonists (H2RAs)

Extremely safe drugs

Adverse Effects

1. Mental status changes (cimetidine)
2. Gynecomastia, impotence in men (cimetidine)



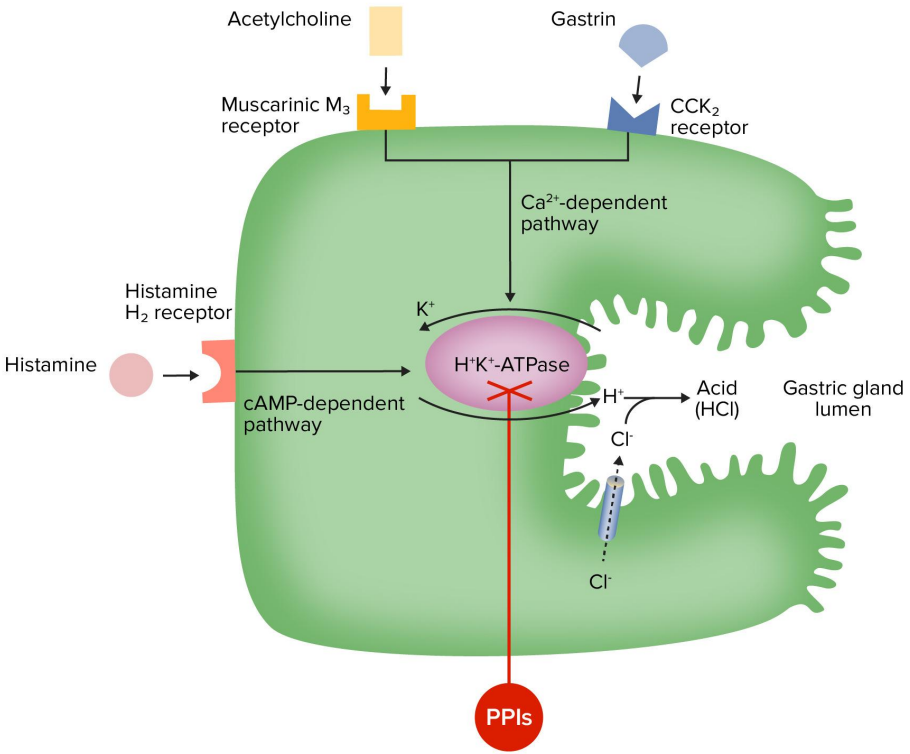
Proton-Pump Inhibitors

Administered as inactive **prodrugs**

Administered approximately **1 hour** before a meal (usually breakfast)

Acid inhibition lasts up to **24 hours** owing to **irreversible inactivation** of the proton pump

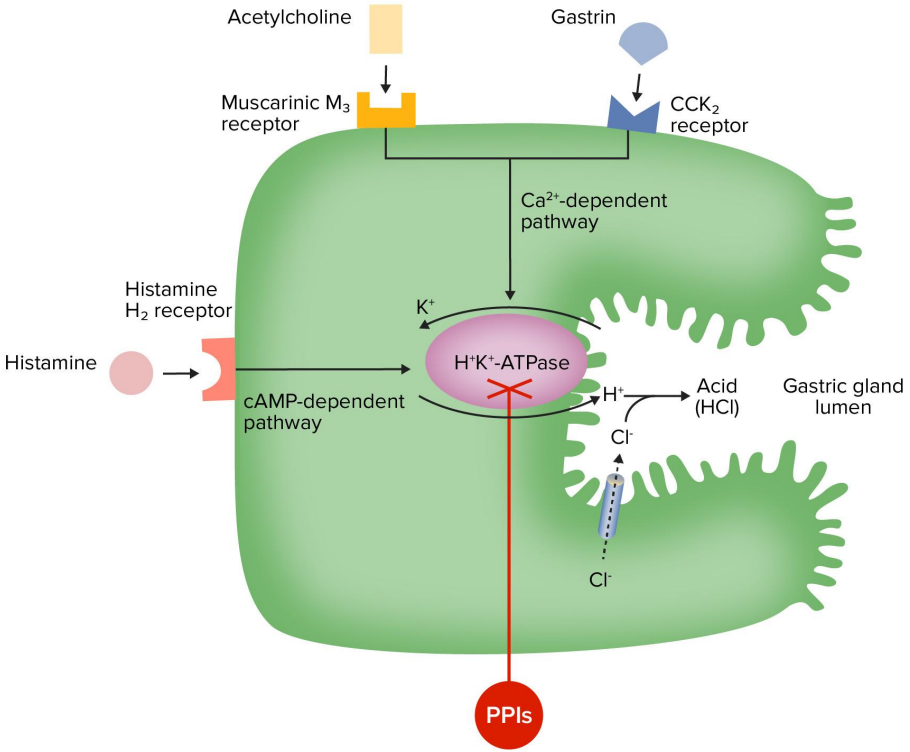
1. Omeprazole
2. Esomeprazole
3. Lansoprazole
4. Dexlansoprazole
5. Rabeprazole
6. Pantoprazole



Proton-Pump Inhibitors

Clinical uses:

1. **GERD**
   - Most effective agents
   - Usually intermittent courses
2. **PUD**
   - more rapid symptom relief and faster ulcer healing than H2RAs
3. **Nonulcer dyspepsia**
4. **Prevention of stress-related mucosal bleeding**
5. **Gastrinoma**

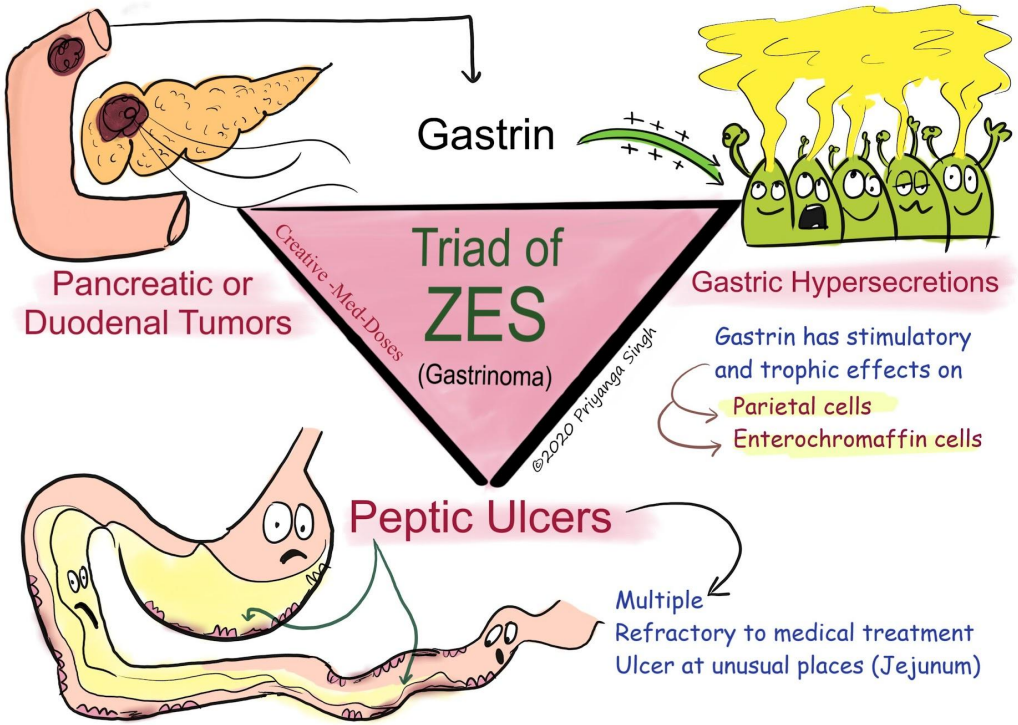


Helicobacter pylori Regimens

H pylori Treatment Regimens

Therapy	Dosing	Duration (days)[a]	Eradication (%)
	First-Line Options		
Clarithromycin triple	PPI[b] bid Clarithromycin 500 mg bid Amoxicillin 1 g bid *or* Metronidazole 500 mg tid	14	70-85
Bismuth quadruple[c]	PPI bid Bismuth subcitrate 120-300 mg/ bismuth subsalicylate 300 mg qid Metronidazole 250 or 500 mg qid Tetracycline 500 mg qid	10-14	75-90
Nonbismuth-based quadruple	PPI bid Amoxicillin 1 g bid Clarithromycin 500 mg bid Metronidazole 500 mg bid	10-14	90

Gastrinoma


Gastrin
Triad of
ZES
(Gastrinoma)
Pancreatic or
Duodenal Tumors
Gastric Hypersecretions
Creative-Med-Doses
©2020 Priyanga Singh
Gastrin has stimulatory
and trophic effects on
Parietal cells
Enterochromaffin cells
Peptic Ulcers
Multiple
Refractory to medical treatment
Ulcer at unusual places (Jejunum)

Proton-Pump Inhibitors

Adverse Effects:

1. Vitamin B12 deficiency
2. Increased respiratory and enteric infections



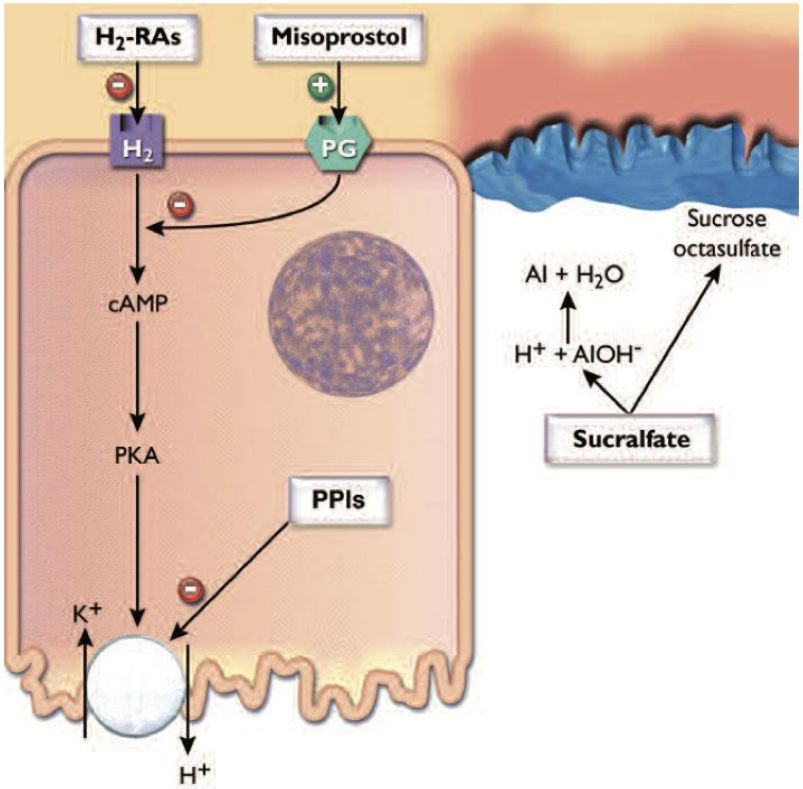


Sucralfate

Salt of sucrose complexed to sulfated aluminum hydroxide

Forms **viscous, tenacious paste** that binds selectively to ulcers or erosions for up to **6 hours**

Administered 1g 4x daily on an empty stomach (1 hr before meals)

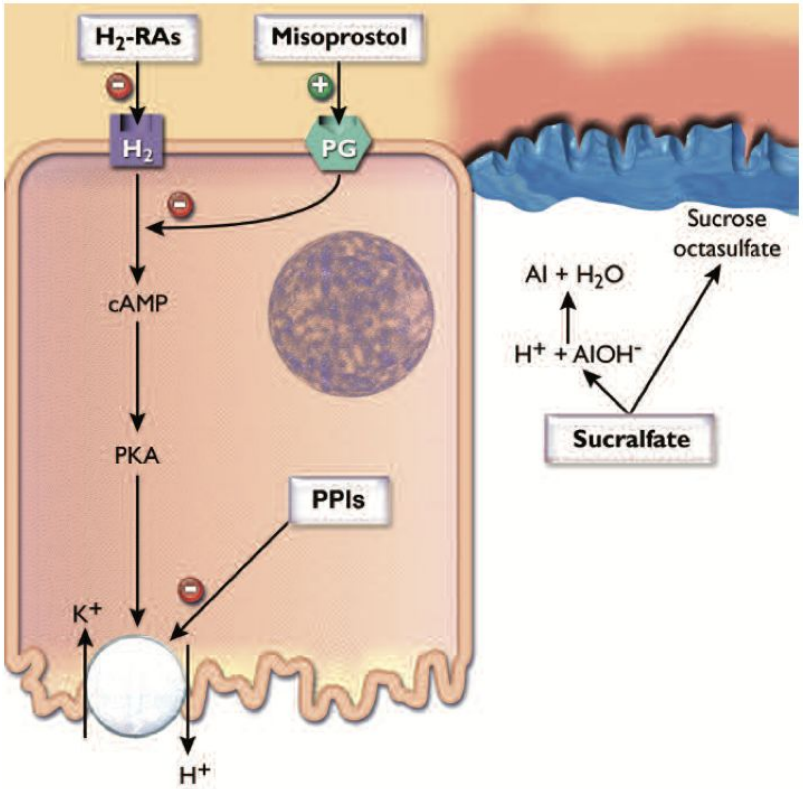


Prostaglandin Analogs

Misoprostol (analog of PGE2)

Administered 3-4x daily

Can reduce incidence of NSAID-induced ulcers

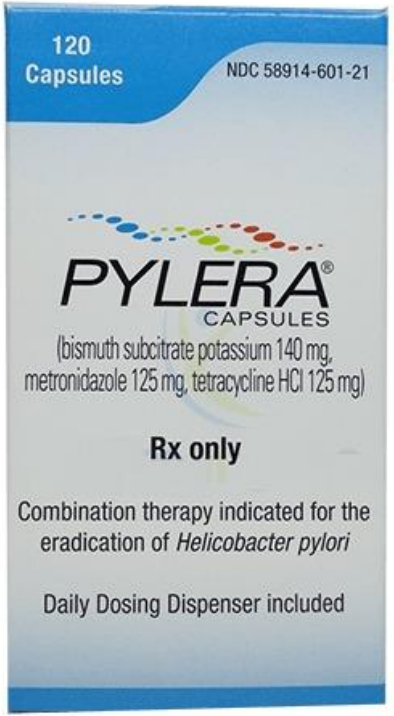


Bismuth Compounds

Coats ulcers and erosions

Stimulate prostaglandin, mucus, and bicarbonate secretion

Has direct antimicrobial activity against H. pylori



Laxatives

Laxatives

1. Bulk-forming laxatives
2. Stool softeners
3. Osmotic laxatives
4. Stimulant laxatives

Bulk-Forming Laxatives

Fiber supplements

Absorb water

Forming a **bulky, emollient gel** that **distends the colon and promotes peristalsis**

1. Psyllium
2. Methylcellulose
3. Polycarbophil

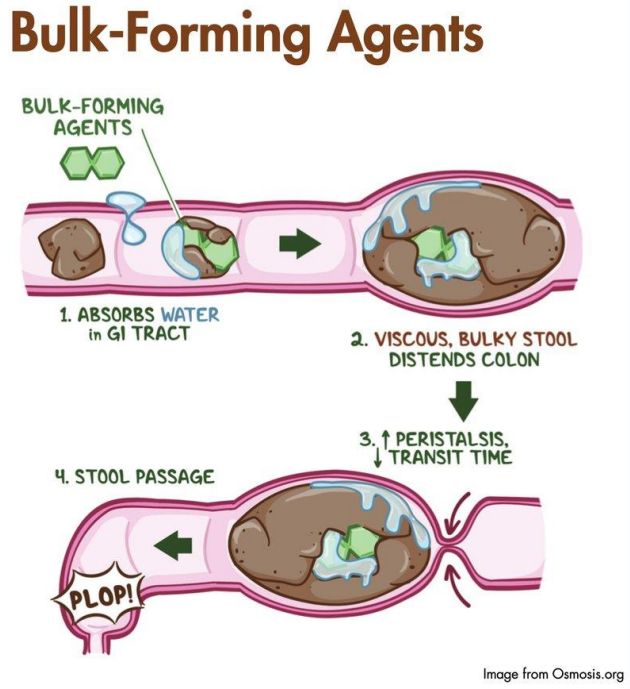


Image from Osmosis.org

Active Ingredient
Psyllium husk; indigestible, fibrous compounds derived from plants

How it Works?
- These agents are not absorbed by the intestines and result to 'bulking up' of the stool
- Softens and increases the bulk of digested food, resulting to colonic distention and increased peristalsis until stools are passed out
- Generally produces effect in 12-72 hours

Considerations
- Can ferment in the intestine, which may cause gas and bloating
- Take with fluids for bulk-forming agents to work and to avoid fecal impaction


American Diabetes Association
PROUD PARTNER
Better Choices for Life
Meta
MUCIL
3 - IN - 1
DAILY FIBER
PSYLLIUM
HUSK FIBER
Helps support: Digestive Health*
Heart Health by Lowering Cholesterol†
Healthy Blood Sugar Levels*
300 CAPSULES

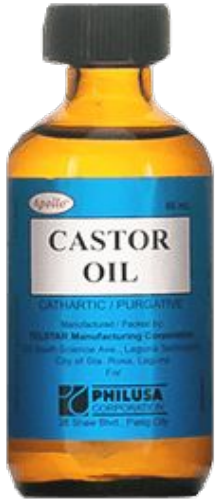
Stool Softeners

Soften stool material, permitting water and lipids to penetrate

May be administered orally or rectally

1. Docusate (oral or enema)
2. Glycerin suppository
3. Mineral oil

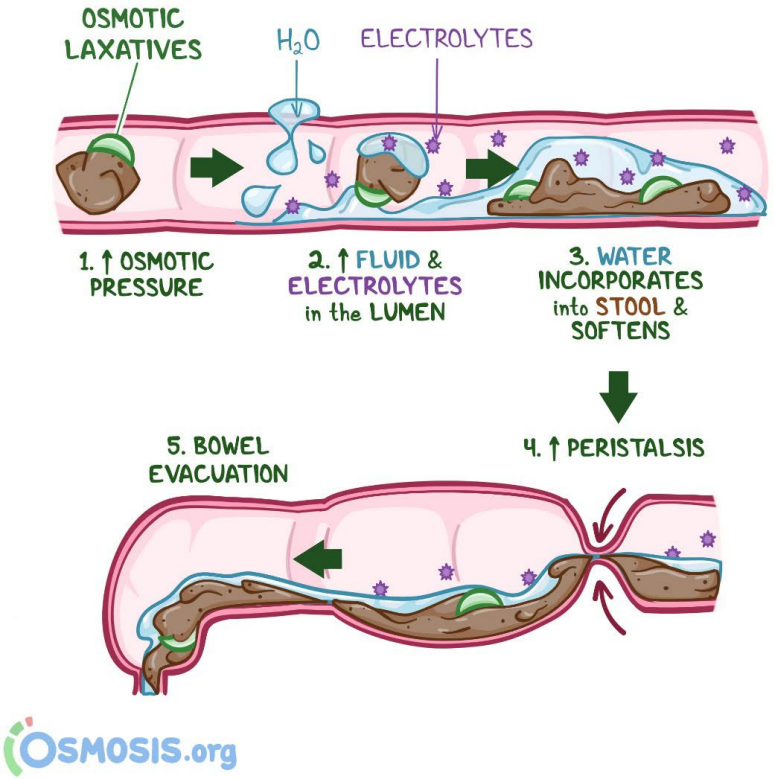




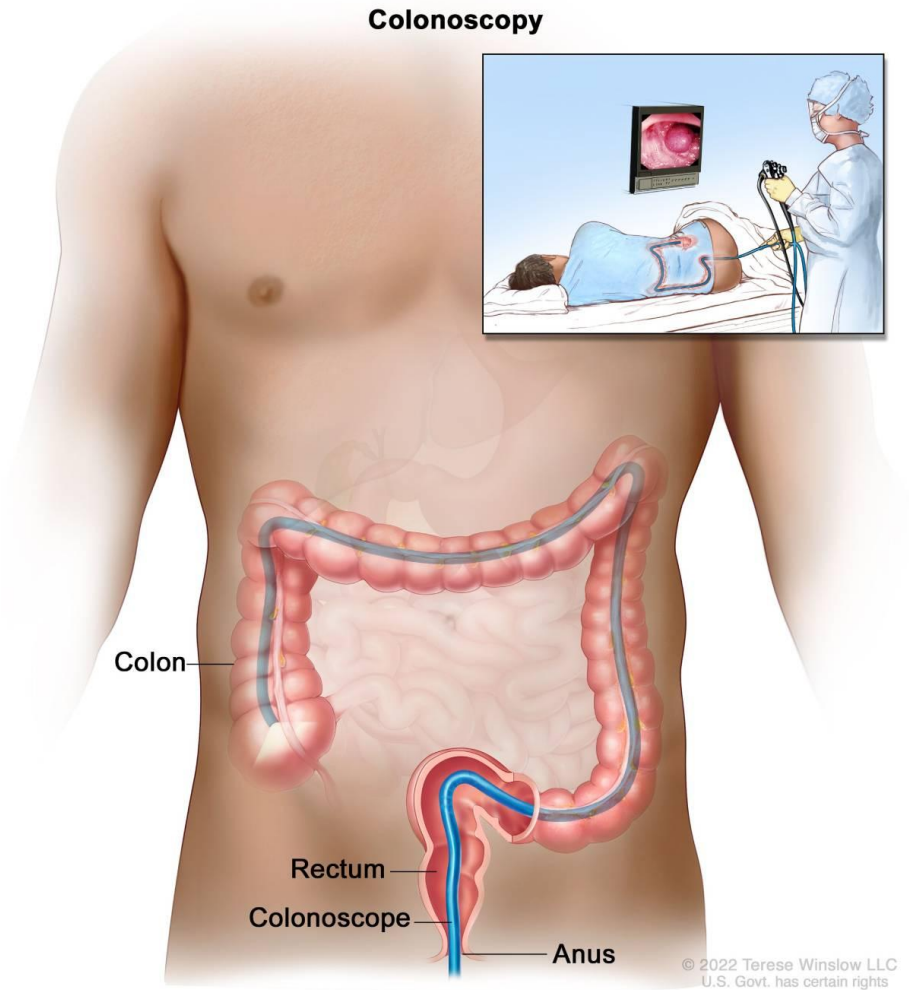
Osmotic Laxatives

Soluble but nonabsorbable compounds that result in **increased stool liquidity due to an obligate increase in fecal fluid**

1. Lactulose
2. Polyethylene Glycol (PEG)
    - used for complete colonic cleansing before GI endoscopy




7 Sachets x 17 g
(Net Wt. 119 g)
Polyethylene Glycol 3350
SureLax®
17 g Powder for Oral Solution
Laxative
Net Wt. 17 g
Polyethylene Glycol 3350
SureLax®
17 g Powder for Oral Solution
Laxative


Colonoscopy
Colon
Rectum
Colonoscope
Anus
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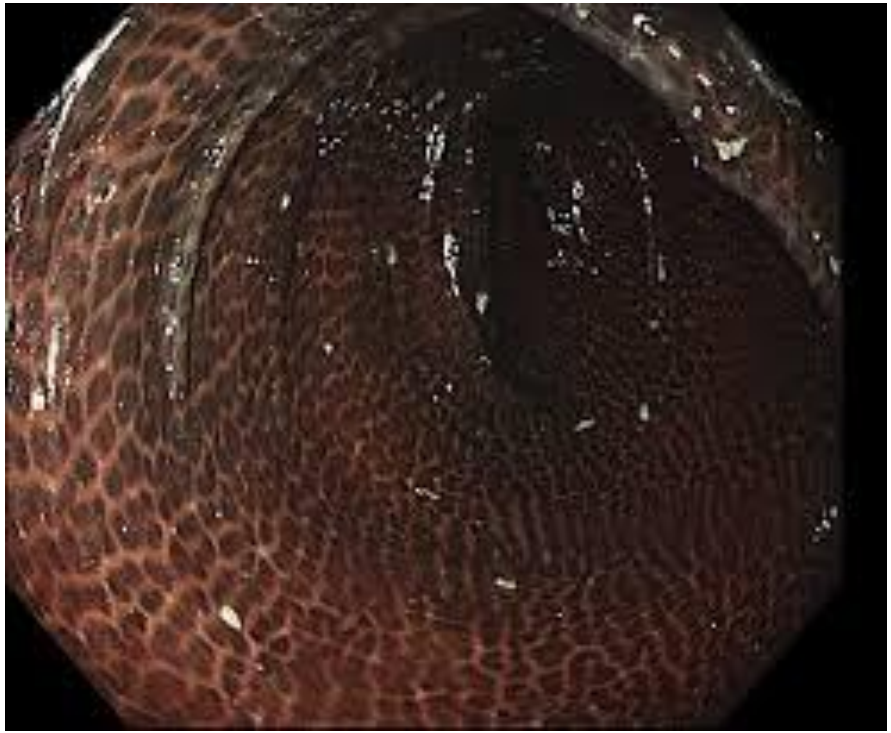
Stimulant Laxatives

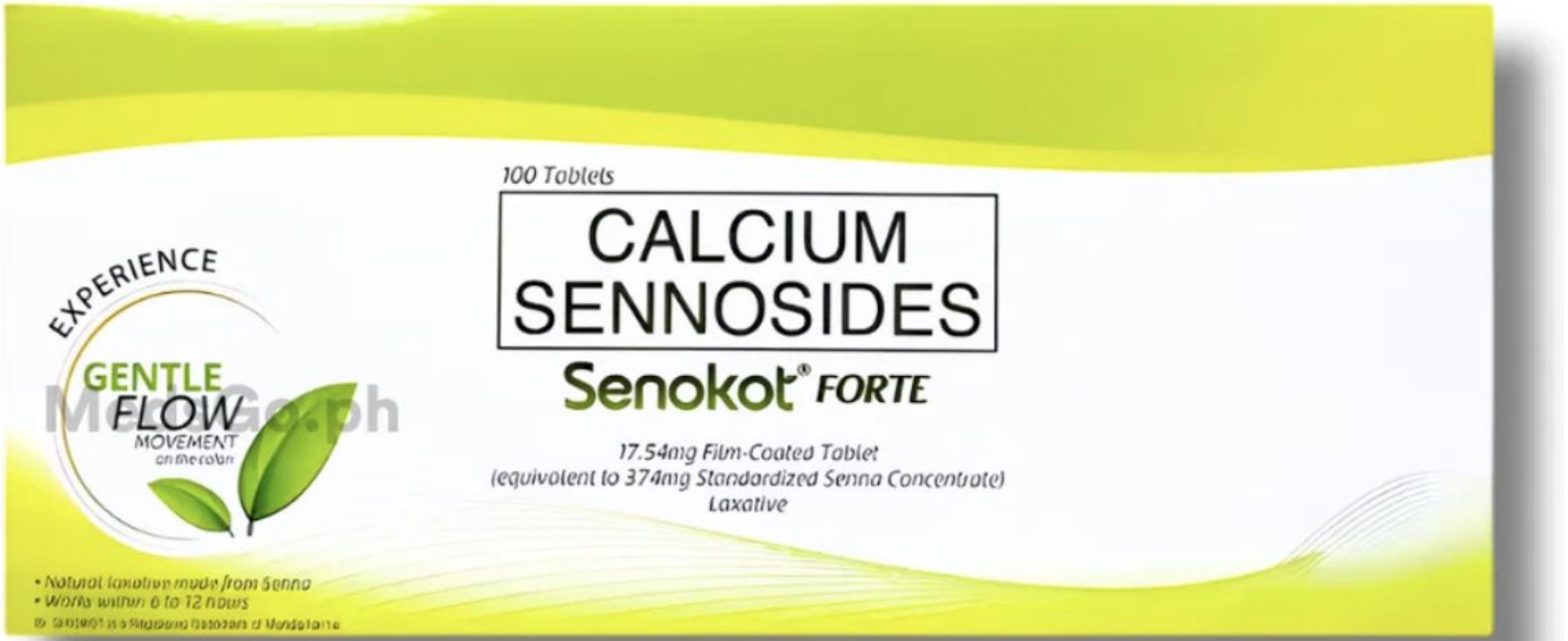
Directly stimulate or irritate the GI lining

1. **Anthraquinone derivatives**
   - Aloe, senna, cascara
   - cause “**melanosis coli**”
2. **Diphenylmethane derivatives**
   - bisacodyl
   - induces bowel movement
     - 6-10hrs (orally)
     - 30-60mins (rectally)




30 Teabags
SENNA LEAVES
SENNA PODS
BIOFITEA
1.275g / 0.425g
LAXATIVE
Manufactured by:
DIAMOND LABORATORIES, INC.
Exclusively
Distributed by:
REJ DIAMOND
PHARMACEUTICALS INC




100 Tablets
CALCIUM SENNOSIDES
Senokot® FORTE
17.54mg Film-Coated Tablet
(equivalent to 374mg Standardized Senna Concentrate)
Laxative
EXPERIENCE
GENTLE
FLOW
MOVEMENT
on the colon
• Natural laxative made from Senna
• Works within 6 to 12 hours

Thank you!